China Economic Review 90 (2025) 102367

Contents lists available at ScienceDirect

China Economic Review

journal homepage: www.elsevier.com/locate/chieco

"A tale of two rails": Transportation infrastructure and technological spillovers from R&D center foreign firms

Lei Li [a], Bocong Liu [b], Bin Sheng [b], Tianyu Wang [c,1,*]

[a] Center for Transnationals' Studies, Economic Behavior and Policy Simulation Laboratory, Nankai University, 94 Weijin Road, Nankai District, Tianjin 300071, China
[b] School of Economics, Nankai University, 94 Weijin Road, Nankai District, Tianjin 300071, China
[c] Institute of World Economy, Jiangsu Provincial Academy of Social Sciences, 168 Jianye Road, Qinhuai District, Nanjing 210004, China

ARTICLE INFO

JEL codes:
F21
R40
O31

Keywords:
Transport infrastructure
R&D center foreign firm
Technology spillover effect
High-speed rail
Subway

ABSTRACT

Identifying the pure technology spillover effect of foreign investment is an academic problem. The academic community also knows little about the role of transportation infrastructure in it. For the first time, this article uses the special event of R&D center foreign firms entering to explore how two completely different transportation infrastructures, high-speed rail, and subway, promote technology spillover from foreign investment within the same empirical framework. Based on city-industry-year panel data in China from 2004 to 2020, this article empirically proves that both high-speed rail and subway can promote R&D center foreign firms' technology spillover. Further analysis found that transportation infrastructure and R&D center foreign firms can jointly promote the development of China's key industries, that is, the core effect is stronger for high-tech and strategic emerging industries. This article also provides evidence about cross-city technology spillover from R&D center foreign firms. First, this article takes advantage of the special nature of R&D center foreign firms to provide new ideas for subsequent related research. Second, it empirically confirms that transportation infrastructure can promote foreign investment technology spillovers.

1. Introduction

Developing countries commonly regard attracting foreign direct investment (FDI) as a crucial component of their economic development strategies. FDI has the potential to significantly narrow the productivity gap between developed and developing countries, increase employee wages, and enhance the performance of domestic firms (Abebe, McMillan, & Serafinelli, 2022; Keller, 2004; Setzler & Tintelnot, 2021). The role of transnational technology transfer primarily facilitated through FDI, is particularly vital in this context. Existing research often constructs various FDI penetration indicators at the regional and industry levels or identifies FDI policy shocks to explore the spillover effects of FDI on host countries (Javorcik, 2004; Javorcik & Spatareanu, 2008; Lu, Tao, & Zhu, 2017; Poole, 2013; Setzler & Tintelnot, 2021). However, measuring the scale and penetration of FDI at the regional and industry levels, or utilizing specific FDI policies, might lead to confounding issues when identifying specific types of FDI spillover effects. For instance,

* Corresponding author.
E-mail addresses: nklilei@nankai.edu.cn (L. Li), 1120211112@mail.nankai.edu.cn (B. Liu), shengbin@nankai.edu.cn (B. Sheng), wangtianyu@jsass.org.cn (T. Wang).
[1] Present/permanent address: 168 Jianye Road, Qinhuai District, Nanjing 210004, China

https://doi.org/10.1016/j.chieco.2025.102367
Received 26 August 2024; Received in revised form 22 November 2024; Accepted 9 February 2025
Available online 14 February 2025
1043-951X/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

the entry of foreign-invested firms into a region or industry not only brings about technological spillovers but may also induce market competition effects on domestic firms. Thus, distinguishing between the technological spillover effects and market competition effects of FDI to obtain a pure estimate of technological spillovers has emerged as a significant academic challenge.

To address the aforementioned issue, some studies have taken alternative approaches by utilizing specific modes of FDI to identify spillover effects. For example, Alfaro-Urena, Manelici, and Vasquez (2022) examined the impact of Costa Rican firms joining the supply chains of multinational corporations (MNCs). They found that these local firms experienced significant improvements in employment, total factor productivity (TFP), and long-term sales after supplying to MNCs. Gong (2023), on the other hand, cleverly exploited the technological linkages between U.S. publicly traded companies and their Chinese subsidiaries, revealing that innovations by U.S. companies significantly enhanced the productivity of Chinese domestic firms located in the same county as the subsidiaries.

This paper builds on existing literature and proposes the use of R&D center foreign firms to identify technology spillover effects from FDI, contributing to the current body of research. The advantages of this approach are twofold: First, innovation and R&D are one of the primary responsibilities of R&D center foreign firms. Compared to regular foreign firms, the production and sales of products by R&D center foreign firms represent a smaller proportion of their operations (supporting evidences will be provided in Section 2 and the Appendix), thus reducing the potential interference of FDI on the market competition effects of host country firms. Second, unlike mainstream business-oriented foreign firms, the inherent characteristics of R&D center foreign firms ensure that they have stronger technology spillover effects on the host country, which facilitates more accurate estimation of technology spillovers in empirical studies.

In addition to the academic issue of identifying the technology spillover effects of FDI, there is another practical question that needs to be addressed: Even if FDI does lead to technology spillovers, it is not guaranteed that the host country will benefit from such spillovers in terms of innovation activities. A representative example is the case of poor transportation infrastructure, which can hinder the diffusion and adoption of technology. For instance, Gebresilasse (2023) found that in Ethiopia, agricultural productivity could only be significantly improved when rural roads were integrated with agricultural extension programs.

In the context of this study, on the one hand, inadequate transportation infrastructure limits the ability of host country firms to access foreign-invested institutions, thereby weakening the spatial diffusion of technology spillovers. The decay of technology spillovers over spatial distance is substantial (Giroud, Lenzu, Maingi, & Mueller, 2024). Thus, good transportation infrastructure can enhance technology spillovers. For example, the establishment of Sweden's national railway network reduced interaction costs, facilitating patent collaborations among inventors and the formation of a national market for ideas (Andersson, Berger, & Prawitz, 2023; Berger & Prawitz, 2024). On the other hand, if poor transportation infrastructure limits the ability of host country firms to access external markets, then even if there are incentives for technology spillovers from FDI, host country firms may be unwilling to invest in high-risk innovation activities due to the lack of market opportunities to translate technology into profits. For example, Suri (2011) found that when access to knowledge and inputs improved, farmers would only invest in costly modern inputs if they could also enter output markets, making such investments profitable. A natural extension of the above analysis is the question of whether improvements in regional transportation infrastructure would enhance the technology spillover effects of FDI. This issue is crucial for China and many developing countries, as it concerns whether they can absorb more international advanced technology by simultaneously promoting FDI inflows and developing transportation infrastructure, thus becoming an urgent practical concern.

In addressing the above issue, this paper examines how two distinct types of transportation infrastructure—high-speed rail (HSR) and subway—amplify the technology spillover effects of R&D center foreign firms within the same empirical framework: the former promotes inter-regional communication, while the latter enhances intra-regional communication. Specifically, the HSR network improves regional accessibility, reduces inter-regional commuting times (Cao, Liu, Wang, & Li, 2013; Jiao, Wang, Jin, & Dunford, 2014; Liu & Zhang, 2018; Shaw, Fang, Lu, & Tao, 2014), expands the range of travel, and lowers communication costs between regions. This suggests that HSR can reduce information asymmetry and uncertainty when firms access external markets, thereby promoting inter-regional connections and economic integration (Borusyak & Hull, 2023; Chen & Haynes, 2017; Jiao et al., 2014). In this process, innovative entities can achieve greater market returns through the HSR network on a national scale, rather than being confined to their local city. Studies by Donaldson and Hornbeck (2016) and Donaldson (2018) show that railway networks can combine local and external markets, increasing market demand for innovation on one hand, and reducing input costs in the innovation process by facilitating regional exchanges on the other.

The opening of subway alleviates urban traffic congestion, reduces intra-regional commuting times (Anderson, 2014; Gu, Jiang, Zhang, & Zou, 2021; Severen, 2023; Yang, Chen, Qin, Lu, & Liu, 2018), and lowers the communication costs of interactions between entities within the same region, thereby enhancing the likelihood or efficiency of host country entities absorbing technology spillovers from R&D center foreign firms. In fact, subways shorten travel times, reduce communication costs, and increase opportunities for interaction, thereby facilitating knowledge diffusion, such as the absorption of knowledge from more distant areas and promoting collaborative innovation (Koh, Li, & Xu, 2022; Wang, Liu, Zhang, & Hu, 2022). The above evidence suggests that subways play an important role in promoting intra-regional exchanges: subways stimulate communication between different entities, spread ideas and experiences, and thus amplify the technology spillover effects brought by R&D center foreign firms.

This paper first conducts an empirical analysis using city-industry-year panel data in China from 2004 to 2020, finding that the entry of R&D center foreign firms significantly boosts innovation levels within the same city-industry combination, thereby confirming the existence of technology spillover effects from R&D center foreign firms. It then intersects binary variables for the presence of HSR and subway with binary variables for the entry of R&D center foreign firms in city-industry combinations. The empirical results show that the coefficients for both interaction terms are significantly positive, demonstrating that both HSR and subway infrastructure amplify the technology spillover effects from R&D center foreign firms. Further analysis reveals that the amplification effect of transportation infrastructure on technology spillovers is stronger in regions with higher marketization levels and better rule-of-law

environments. Additionally, the paper finds that R&D center foreign firms can generate technology spillovers across cities, meaning transportation infrastructure also amplifies spillovers from R&D center foreign firms in surrounding cities and industries. Finally, the amplification effect of transportation infrastructure on technology spillovers is stronger in high-tech and strategic emerging industries, providing evidence that both transportation infrastructure and R&D center foreign firms contribute to the development of key industries in China.

The potential marginal contributions of this paper are as follows: First, it is the first to identify technology spillover effects from FDI based on the entry of R&D center foreign firms, effectively reducing the interference of market competition effects, thus addressing the theoretical issue of identifying FDI technology spillovers. Existing studies that focus on the scale, penetration, and policies related to FDI at the regional or industry level (e.g., Javorcik, 2004; Javorcik & Spatareanu, 2008; Lu et al., 2017; Poole, 2013; Setzler & Tintelnot, 2021) struggle to differentiate between the market competition effects and technology spillovers of foreign-invested enterprises on host country firms, making it difficult to estimate pure FDI technology spillovers. By leveraging the unique nature of R&D center foreign firms—whose primary responsibility is innovation and R&D—this paper effectively avoids the market competition typically associated with business-oriented foreign firms and isolates the pure technology spillovers from R&D center foreign firms. Second, while existing research primarily focuses on the spillover effects of transportation infrastructure itself (e.g., Cantos, Gumbau-Albert, & Maudos, 2005; Marinos, Belegri-Roboli, Michaelides, & Konstantakis, 2022; Yu, De Jong, Storm, & Mi, 2013), less attention has been paid to how transportation infrastructure influences the spillover effects generated by other entities. This paper, within the same empirical framework, examines and finds that both HSR and subway amplify the technology spillover effects of R&D center foreign firms. In doing so, it not only enriches the literature on the effects of transportation infrastructure but also addresses the practical question of whether the simultaneous promotion of FDI inflows and transportation infrastructure development can absorb more international advanced technological experience. Third, the study of how transportation infrastructure amplifies the technology spillovers from R&D center foreign firms helps raise awareness in society that foreign investment requires adequate external conditions to produce significant spillover effects. It also contributes to the formulation of sound and reasonable policies related to foreign investment attraction, innovation R&D, and transportation infrastructure planning at various levels of government.

The remainder of the paper is structured as follows. Section 2 presents the theoretical analysis. Section 3 outlines the research design, including the data used, the empirical model, and the identification strategy. Section 4 verifies the existence of technology spillover effects from R&D center foreign firms and shows that both HSR and subway significantly amplify these spillover effects. Section 5 discusses the mechanism analysis. Section 6 provides further analysis. Section 7 concludes the paper.

2. Theoretical hypotheses

From a cost-benefit perspective, firms will choose to innovate if the innovation benefits exceed the innovation costs. Therefore, external shocks that can increase innovation benefits or reduce innovation costs will promote regional innovation levels. In the context of this study, FDI, represented by R&D center foreign firms, can facilitate technology diffusion, such as host country firms observing and imitating foreign firms, absorbing related management, business, and R&D knowledge (Abebe et al., 2022; Gong, 2023). As such, the entry of R&D center foreign firms can lower innovation costs, relax the R&D constraints faced by host country firms, and reduce the technological barriers to innovation.

How does transportation infrastructure amplify the technology spillover effects from R&D center foreign firms? The research by Gebresilasse (2023) provides useful insights. They examined the joint impact of agricultural extension projects and rural roads on agricultural productivity in Ethiopia. They found that while each factor individually had no effect, their combined presence significantly improved agricultural productivity. This was because the agricultural extension project and rural roads jointly relaxed two interdependent rural constraints: market access and technology acquisition.

The mechanisms through which HSR and subway amplify technology spillovers from R&D center foreign firms differ due to their distinct transportation functions: HSR facilitates inter-regional communication, while subway promotes intra-regional communication. Specifically, HSR networks enhance regional accessibility, reduce inter-regional commuting time (Cao et al., 2013; Jiao et al., 2014; Liu & Zhang, 2018; Shaw et al., 2014), expand the travel range, and lower communication costs between regions. This means that HSR can reduce information asymmetry and uncertainty when firms access markets in other regions, thereby promoting inter-regional connections and economic integration (Borusyak & Hull, 2023; Chen & Haynes, 2017; Jiao et al., 2014). In this process, innovative entities can obtain larger market returns nationwide through the HSR network, rather than being limited to their own region. As a result, HSR relaxes the market access restrictions faced by firms, increases the potential returns to innovation, and amplifies the technology spillover effects from R&D center foreign firms. As Acemoglu and Linn (2004) point out, the larger the market size, the higher the expected return on innovation investment. Research by Donaldson and Hornbeck (2016) and Donaldson (2018) suggests that railway networks can integrate local and external markets, increasing market demand for innovation.

The opening of subway alleviates urban traffic congestion, reduces intra-regional commuting time (Anderson, 2014; Gu et al., 2021; Severen, 2023; Yang et al., 2018), lowers communication costs between local entities, and thus enhances the probability or efficiency with which host country firms absorb technology spillovers from R&D center foreign firms within the same region. Therefore, subways relax the technical acquisition constraints faced by firms, reducing the potential costs of innovation and amplifying the technology spillover effects from R&D center foreign firms. In fact, subways reduce travel time, lower communication costs, and increase interaction opportunities, facilitating knowledge dissemination, such as absorbing knowledge and engaging in cooperative innovation from more distant locations (Koh et al., 2022; Wang et al., 2022). The evidence above shows that subways play an important role in promoting intra-regional communication: subways stimulate interaction between different entities, spread ideas and experiences, and thus amplify the technology spillover effects from R&D center foreign firms.

The analysis above indicates that for external shocks to stimulate innovation, they must lead to a sufficient increase in innovation benefits or a reduction in innovation costs. Fig. 1 illustrates this point. In the context of HSR, subway openings, and the entry of R&D center foreign firms, the entry of R&D center foreign firms implies certain technology spillovers, which reduce the R&D costs of host country firms. Under the condition of R&D center foreign firm entry, the opening of HSR relaxes the market access restrictions faced by firms, enhancing the potential benefits of innovation and thereby amplifying the technology spillover effects from R&D center foreign firms (case 1). Under the condition of R&D center foreign firm entry, the opening of the subway reduces communication costs between local entities, improves the efficiency of technology spillovers from R&D center foreign firms within the same region, relaxes the technical acquisition constraints faced by local firms, and further reduces the potential costs of innovation (case 3). Since this paper focuses on how transportation infrastructure amplifies technology spillover effects from R&D center foreign firms, case 2 falls outside the scope of the study and is not discussed further.

3. Research design

3.1. Data

The core processing variable of this study is the indicator of the entry of R&D center foreign firms at the city-industry level. To obtain this information, we used the business registration data from Qichacha.[2] Specifically, we utilized Qichacha's advanced search feature to identify all firms with "R&D center" in their name or description,[3] and then filtered these by type to select foreign-invested firms, thereby constructing a sample of R&D center foreign firms. The business registration information includes the establishment date, address, and industry of the firms. Based on this data, we created a binary variable indicating the presence or absence of R&D center foreign firms at the city-industry level. Indeed, existing literature has demonstrated the feasibility of using business query websites like Qichacha and Tianyancha to identify specific types of firms. For instance, Beraja et al. (Beraja, Kao, Yang, & Yuchtman, 2023; Beraja, Yang, & Yuchtman, 2023) used Tianyancha to collect data on all facial recognition AI companies in China, which provides confidence in the data collection method employed in this study.

The city-level innovation indicators in this study are derived from the "China City and Industry Innovation Report" by Kou and Liu (2017). This report, based on the Chinese patent database, measures the innovation index at the city-industry level from 2004 to 2020. Information on HSR openings comes from the CNRDS database, which includes details such as opening dates, city locations, and routes of each HSR station. Using this data, a binary variable indicating the presence or absence of HSR in cities was constructed. Information on subway openings is primarily sourced from the "China Urban Construction Statistical Yearbook," specifically the section on urban rail transit construction. Based on this data, a binary variable indicating the presence or absence of subway in cities was created. City statistical data were mainly obtained from the "China City Statistical Yearbook." City transportation data, including metrics such as road area, number of buses, and number of taxis, come from the CEIC database. For missing data, averages from adjacent periods were used for imputation. All relevant variables were deflated using 2000 as the base year.

3.2. Empirical model

The aim of this paper is to test whether two fundamentally different types of transportation infrastructure, HSR and subway, can amplify the technology spillover effects from R&D center foreign firms. First, this paper constructs empirical model (1) to demonstrate the existence of technology spillover effects from R&D center foreign firms. Then, following the approach of Gebresilasse (2023), this paper constructs empirical model (2) to examine whether HSR and subway can amplify the technology spillover effects from R&D center foreign firms.

$$lninnovation_{ift} = \alpha + \beta frdc_{igt} + \delta \mathbf{X}_{igt} + \theta_{if} + \mu_{it} + \tau_{ft} + \varepsilon_{ift} \tag{1}$$

$$lninnovation_{ift} = \alpha + \beta_1 hsr_{it} \times frdc_{igt} + \beta_2 subway_{it} \times frdc_{igt} + \gamma_1 hsr_{it} + \gamma_2 subway_{it} + \gamma_3 frdc_{igt} + \delta \mathbf{X}_{igt} + \theta_{if} + \mu_{it} + \tau_{ft} + \varepsilon_{ift} \tag{2}$$

In these models, i denotes cities, and t denotes years; f represents gb4 industries and g represents gb2 industries.[4] Given that the "China City and Industry Innovation Report" by Kou and Liu (2017) provides the innovation index[5] at the city-gb4 industry level, the dependent variable $innovation_{ift}$ is the innovation index for city i in gb4 industry f in year t, measured to capture city-industry innovation levels and adjusted by taking the natural logarithm of the index plus one. hsr_{it} is a binary variable indicating whether HSR is operational in city i in year t; $subway_{it}$ is a binary variable indicating whether subway is operational in city i in year t; and $frdc_{igt}$ is a

[2] https://www.qcc.com/

[3] Such as "R&D Center", "Research and Development Center", etc.

[4] In this study, gb2, gb3, and gb4 industries refer to the two-digit, three-digit, and four-digit industries of the 2017 National Economic Industry Classification.

[5] Kou and Liu (2017) used invention patents from the Chinese patent database. They identified the city to which each patent belonged and, using the IPC classification codes and the National Economic Industry Code matching table provided by the National Bureau of Statistics of China, allocated the patents to industry sectors. They standardized the total patent value in China for 2001 to 100 and then calculated the innovation index at the city-industry level. The innovation index developed by Kou and Liu (2017) measures the innovation scale at the city-industry level based on invention patents.

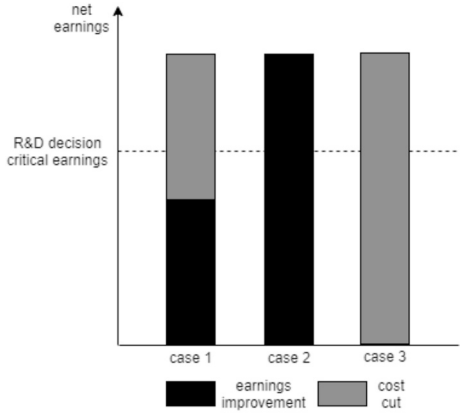

Fig. 1. Description of R&D decisions under different cases.

binary variable indicating the presence of R&D center foreign firms in gb2 industry g in city i in year t.[6]

The core coefficient of interest in eq. (1) is $frdc_{igt}$, with coefficient β. If there is technology spillovers effect from R&D center foreign firms—then $\beta > 0$.The core coefficients of interest in eq. (2) are the interaction terms: $hsr_{it} \times frdc_{igt}$, with coefficient β_1, and $subway_{it} \times frdc_{igt}$, with coefficient β_2. Following the explanation provided by Gebresilasse (2023), if there are innovation complementarities between HSR, subway, and R&D center foreign firms—i.e., if HSR and subway enhance technology spillovers from R&D center foreign firms—then $\beta_1 > 0$ and $\beta_2 > 0$. Additionally, in line with Gebresilasse (2023), the study performs parallel trend and placebo tests on the core interaction terms $hsr_{it} \times frdc_{igt}$ and $subway_{it} \times frdc_{igt}$.

The vector $\mathbf{X}_{igt}$ represents a set of control variables. To account for potential interference from other R&D center firms and foreign firms, this study controls for the logarithm of the number of active R&D center firms at the city-gb2 industry level ($lnrdcnum_{igt}$), and the logarithm of the number of active foreign firms at the city-gb2 industry level, calculated from the complete business registration data of foreign-invested firms from Qichacha ($lnforeignnum_{igt}$). Appendix A provides a detailed explanation of the rationale for selecting these control variables. Since the dependent variable in this study involves three dimensions—city, industry, and year—this allows for very detailed fixed effects controls, including city-gb4 industry fixed effects θ_{if}, city-year fixed effects μ_{it}, and gb4 industry-year fixed effects τ_{ft}. Appendix A elaborates on the contribution of these fixed effects to causal inference. ε_{ift} represents the random error term. Finally, unless otherwise noted, all regressions in this study use robust standard errors clustered at the city-gb2 industry level, as detailed in the table notes. Appendix Table 2 presents the descriptive statistics of the main variables.

3.3. Discussions

The core argument of this paper is that the market competition effect of R&D center foreign firms is weaker, which helps to better identify the pure technology spillover effect of foreign investment. This section provides some evidence and discusses the potential drawbacks of this approach. A direct and effective method would be to obtain annual financial data for all firms and examine whether R&D center foreign firms are engaged in product sales or assess the proportion of their sales scale. However, to our knowledge, there is no comprehensive firm-level financial database in China that covers all firms and the period of this study (2004–2020). As a result, we turn to two large-scale firm databases: the China Industrial Enterprise Database (1998–2013) and the China Tax Survey Database (2007–2016). The China Industrial Enterprise Database covers all state-owned enterprises in industrial sectors and non-state-owned enterprises above a certain scale. The China Tax Survey Database is a survey-based database, with many firms entering and exiting the survey each year. This paper uses firm names to strictly match and fuzzy match R&D center foreign firms with both the China Industrial Enterprise Database and the China Tax Survey Database, calculating the size ratio of R&D center foreign firms and other foreign firms within the same city-gb2 industry. Firm size is measured by production, sales, assets, and employment. We find that R&D center foreign firms make up a very small proportion of the total size of all firms. In contrast, other foreign firms account for a much larger share, far exceeding that of R&D center foreign firms. Appendix B and Table B1 provide further details.

The R&D center foreign firms identified in this paper include purely R&D-focused firms as well as those that combine R&D with production and sales activities. Appendix B provides the registration background and specific case studies of these firms in China. The former does not generate potential market competition effects, while the latter may generate both market competition and technology

[6] Appendix A explains why identifying the entry of R&D center foreign firms in the city-gb2 industry level is used in this study. Empirical results presented later indicate that identifying R&D center entry at the city-gb3 and city-gb4 industry levels may lead to an upward aggregation of the dependent variable, i.e., the innovation index at the city-gb3 industry level and city-gb2 industry level, with results remaining robust. For clarity, Appendix TableA1 provides an example based on the automotive manufacturing industry (industry code 36), omitting the city dimension, to illustrate how to construct binary variables for R&D center entry at the gb2, gb3, and gb4 industry levels.

spillover effects. However, compared to regular foreign firms, their technology spillover effects are stronger because they have internal R&D departments rather than focusing solely on production and sales. The regression results provide suggestive evidence (e.g., Column (2) and Column (4) of Table 1). The paper finds that the regression coefficient of the control variable, the logarithm of the number of foreign firms in the city-gb2 industry (*lnforeignnum*), is not significant and has a very small effect. A possible explanation is that the positive foreign investment technology spillover effect is offset by the negative market competition effect. In contrast, the coefficient of the control variable, the logarithm of the number of R&D center firms in the city-gb2 industry (*lnrdcnum*), is significantly positive. This suggests the validity of the method used to collect information on R&D center firms—unlike regular firms, R&D center firms significantly enhance innovation levels within the same city-industry and generate significant technology spillover effects.

Statistical data analysis reveals that, between 2004 and 2020, the number of R&D center foreign firms made up a small proportion of all foreign firms, but there is strong heterogeneity in their industry distribution. This paper groups firms according to the high-tech industry classification provided by the National Bureau of Statistics of China and calculates the Revealed Comparative Advantage Index (RCA) of R&D center foreign firms in high-tech vs. non-high-tech industries. The paper finds that, based on either firm quantity or registered capital, the RCA for R&D center foreign firms in high-tech industries is much higher than in non-high-tech industries, ranging from 2.2 to 2.8 times. This suggests that R&D center foreign firms are more concentrated in high-tech industries compared to regular foreign firms. The paper argues that R&D center foreign firms represent a small but significant proportion of foreign firms in China, and their concentration in high-tech industries could lead to stronger technology spillover effects. Appendix B and Table B2 provide further details.

Lastly, this paper discusses the differences between the treatment group and the control group in the appendix. It examines the differences in innovation levels at the city-gb4 industry level, comparing the mean of the innovation index logarithms between the groups. The paper then compares the differences in urban market size between the treatment and control groups, where urban market size is measured by the logarithm of GDP and total population. The paper finds that, in an unconditional comparison, the treatment group has significantly higher innovation levels and urban market size than the control group. Panel regressions at the city-gb4 industry level and urban level, with control variables and fixed effects, show that after controlling for variables and fixed effects, the difference between the treatment and control groups is much smaller or no longer significant. Appendix B and Table B3 provide further details.

These results suggest that while the treatment group has significantly higher innovation levels and urban market size compared to the control group, the difference is no longer significant or is greatly reduced after controlling for variables and fixed effects. To further test whether the differences between the treatment and control groups in innovation levels and urban market size might interfere with the estimates, the paper will provide robust evidence in Section 4.4.

It is important to note that while this paper provides some evidence supporting that the market competition effect of R&D center foreign firms is weaker than that of regular foreign firms, and uses this characteristic of R&D center foreign firms to identify the pure technology spillover effect of foreign investment, the paper acknowledges three potential limitations that must be considered. First, due to the lack of a comprehensive firm financial database, these pieces of evidence are not direct. Furthermore, financial indicators cannot fully reflect a firm's competitive strength, so caution is needed in interpretation. Second, the use of Qichacha to identify R&D center foreign firms may lead to both over-identification and under-identification issues, which require more precise identification methods in future research. Third, there are still market competition effects associated with R&D center foreign firms. The significant positive impact of R&D center foreign firms on innovation levels, compared to the weak impact of regular foreign firms, may be due to the different reactions of local firms to market competition from R&D center foreign firms versus regular foreign firms: when local firms face market competition from R&D center foreign firms, the escape competition effect is greater than the Schumpeterian effect, leading to increased innovation incentives; when local firms face competition from regular foreign firms, the Schumpeterian effect

Table 1
Baseline regression.

Dependent variable: ln*innovation*variable:*innovation*	(1)	(2)	(3)	(4)
hsr×*frdc*			0.0488***	0.0477***
			(0.00610)	(0.00608)
subway×*frdc*			0.0583***	0.0581***
			(0.00773)	(0.00771)
frdc	0.0285***	0.0206***	−0.0180**	−0.0238***
	(0.00746)	(0.00767)	(0.00706)	(0.00723)
ln*foreignnum*		−0.000319		−0.000102
		(0.000541)		(0.000519)
ln*rdcnum*		0.0167***		0.0137***
		(0.00219)		(0.00200)
City-gb4 industry fixed effects	Yes	Yes	Yes	Yes
City-year fixed effects	Yes	Yes	Yes	Yes
Year-gb4 industry fixed effects	Yes	Yes	Yes	Yes
Observations	3,968,244	3,968,244	3,968,244	3,968,244
R^2	0.196	0.214	0.772	0.772

Note: The values in parentheses represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively. The binary variables for HSR opening *hsr* and subway opening *subway* are city-level time-varying variables, which are absorbed by the city-year fixed effects.

dominates the escape competition effect, reducing innovation incentives. This possible explanation cannot be directly tested in this paper and warrants further investigation in future studies.

Additionally, even though significant technology spillover effects are observed from R&D center foreign firms, it is difficult to generalize that other foreign firms would also generate significant technology spillover effects. First, R&D center foreign firms are independent R&D entities or firms with R&D departments, which can generate stronger technology spillover effects. In contrast, regular foreign firms mainly focus on production and sales, which may not generate significant technology spillover effects. Second, the market competition effect of regular foreign firms may suppress the innovation incentives of local firms through the Schumpeterian effect (Liu, Lu, Lu, & Luong, 2021).

4. Empirical results

4.1. Benchmark results

Table 1, columns (1) and (2), present the regression results based on Eq. (1) to examine the existence of the technology spillover effects of R&D center foreign firms. Column (1) does not control for the logarithm of the number of incumbent R&D center foreign firms at the city-gb2 industry level ($lnrdcnum_{igt}$) or the logarithm of the number of incumbent foreign firms at the city-gb2 industry level ($lnforeignnum_{igt}$). In contrast, column (2) includes these two control variables. Both columns saturate the model with fixed effects: city-gb4 industry fixed effects, city-year fixed effects, and gb4 industry-year fixed effects. The results indicate that the entry of R&D center foreign firms significantly increased the innovation index of the same city-industry by 2 %–3 %. Appendix Fig. C1 provides the event study results corresponding to Eq. (1), replacing the binary variable for R&D center foreign firm entry (*frdc*) in Eq. (1) with a set of dummy variables representing periods before and after the entry. The regression excludes the period $t = -1$ (the period immediately before the entry) as the reference point. The findings show no significant differences between the treatment and control groups before the entry of R&D center foreign firms at the city-gb2 industry level, indicating no significant pre-entry trends. This supports the parallel trends assumption.

Table 1, columns (3) and (4), report the regression results based on Eq. (2) to examine whether HSR and subway amplify the technology spillover effects of R&D center foreign firms. Both core interaction terms remain significantly positive, indicating that both types of transportation infrastructure significantly enhance the technology spillovers from R&D center foreign firms. Column (4) is selected as the benchmark specification for subsequent regressions.

Using the results from Table 1, columns (2) and (4) as benchmarks, this study interprets the economic significance of the core interaction terms following the approach of Gebresilasse (2023). Table 1, column (2) indicates that the entry of R&D center foreign firms increases the city-industry innovation index by approximately 2 %. Meanwhile, the results in Table 1, column (4) show that HSR and subway significantly amplify the technology spillover effects of R&D center foreign firms. In cities with HSR, the innovation index increases by approximately 3 % (=5 %–2 %), and in cities with subway, the innovation index increases by approximately 4 % (=6 %–2 %). It is important to note that the impacts of HSR and subway openings are absorbed by city-year fixed effects. This suggests that HSR amplifies the spillover effects of R&D center foreign firms by 1.5 times, and subway amplify these effects by 2 times. These are

Table 2
High-speed rail and subway jointly enhance the technology spillover effect of R&D center foreign firms.

Dependent variable: ln*innovation*	(1)
hsr×*subway*×*frdc*	0.183*** (0.0261)
hsr×*frdc*	0.0169*** (0.00225)
subway×*frdc*	−0.102*** (0.0245)
frdc	−0.00351 (0.00707)
ln*foreignnum*	−3.31e-05 (0.000501)
ln*rdcnum*	0.0130*** (0.00194)
City-gb4 industry fixed effects	Yes
City-year fixed effects	Yes
Year-gb4 industry fixed effects	Yes
Observations	3,968,244
R^2	0.776

Note: The values in parentheses represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively. The binary variables for HSR opening *hsr* and subway opening *subway* and interaction term *hsr*×*subway* are city-level time-varying variables, which are absorbed by the city-year fixed effects.

substantial and significant improvements.

Finally, in Table 2, columns (3) and (4), the coefficients of the variable *frdc* are negative. However, this does not imply that R&D center foreign firms have negative spillover effects in cities without HSR or subway. First, after introducing interaction terms, the main effect no longer depends solely on its coefficient; it also depends on the coefficients of the interaction terms. Second, the focus of this study is on the causal identification of the interaction terms $hsr \times frdc$ (HSR and R&D center foreign firm entry) and $subway \times frdc$ (subway and R&D center foreign firm entry). Therefore, the *frdc* coefficient's ability to serve as a reliable causal estimate is questionable once interaction terms are included. To further examine this, the study adopts the specification of Eq. (1) and conducts subgroup regressions based on whether a city has HSR or subway. The results are shown in Appendix Table C1. No evidence suggests that R&D center foreign firms have negative spillover effects in cities without HSR or subway. However, caution is required when interpreting subgroup regression results. First, the assumptions underlying interaction term regressions differ from those of subgroup regressions, particularly regarding sample size and the composition of treatment and control groups. Second, only a small number of Chinese cities have subway, significantly reducing the sample size in column (4) and lowering estimation power.

4.2. Parallel trend test

To test whether the two key interaction terms satisfy the parallel trends assumption, this study follows the approach of Gebresilasse (2023) and employs an event study methodology, as detailed below:

$$lninnovation_{ift} = \alpha + \sum_{k=-5}^{5} \beta_k^h impact_{ig,t=k}^{hf} + \sum_{j=-5}^{5} \beta_k^s impact_{ig,t=j}^{sf} + \gamma_1 hsr_{it} + \gamma_2 subway_{it} + \gamma_3 frdc_{igt} + \delta \mathbf{X}_{igt} + \theta_{if} + \mu_{it} + \tau_{ft} + \varepsilon_{ift} \quad (3)$$

In this model, the variable $impact_{ig,t=k}^{hf}$ is a dummy variable that equals 1 if year t is the kth year after the HSR opened and an R&D center foreign firm entered in city i and sector g. Similarly, $impact_{ig,t=j}^{sf}$ is a dummy variable that equals 1 if year t is the jth year after the subway opened and an R&D center foreign firm entered in city i and sector g. Therefore, $impact_{ig,t=k}^{hf}$ measures the time relative to the year when the HSR opened and the R&D center foreign firms entered city i and sector g: if $k < 0$, $impact_{ig,t=k}^{hf}$represents the $|k|$th year before the HSR opening and R&D center foreign firms entry in city i; if $k > 0$, $impact_{ig,t=k}^{hf}$ represents the $|k|$th year after the HSR opening and R&D center foreign firms entry in city i; if $k = 0$, $impact_{ig,t=k}^{hf}$ represents the year when the HSR opened and the R&D center foreign firms entered in city i. The interpretation of $impact_{ig,t=k}^{sf}$ is similar to that of $impact_{ig,t=k}^{hf}$, but with reference to the subway opening instead. Following the approach of Abeberese, Barnwal, Chaurey, and Mukherjee (2023), the analysis includes a five-year period before and after the impact (i.e., $k,j \in [-5, 5]$) for the parallel trends test. The periods before year −5 and after year 5 are aggregated into $impact_{ig,t=-5}^{hf}$and $impact_{ig,t=5}^{hf}$ (as well as $impact_{ig,t=j}^{sf}$). The year immediately before the impact, $k, j = -1$, is excluded from the regression as the reference point. Control variables, fixed effects, and the level of clustering are consistent with those in eq. (2), and all main effects of the two key interaction terms (i.e., the HSR opening dummy variable *hsr*, the subway opening dummy variable *subway*, and the R&D center foreign firms entry dummy variable *frdc*) are included in the regression.

Fig. 2 reports the parallel trend test results, presenting the estimated coefficients and corresponding 95 % confidence intervals for the two key interaction terms from eq. (2). The dots display the results for the HSR opening combined with the entry of R&D center foreign firms, corresponding to the interaction term $hsr \times frdc$. The rhombus present the results for the subway opening combined with

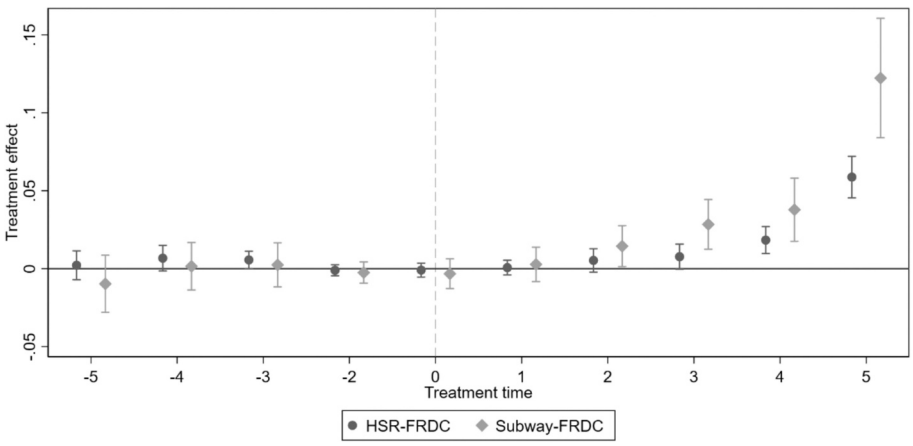

Fig. 2. Parallel trend test results.

Note: Fig. 2 presents the coefficient estimates and corresponding 95 % confidence intervals for the two key interaction terms from the event study specification in eq. (3). The dots show the results for the event where HSR opened and R&D center foreign firms entered, representing the interaction term *hsr*×*frdc*. The rhombus display the results for the event where the subway opened and R&D center foreign firms entered, representing the interaction term *subway*×*frdc*.

the entry of R&D center foreign firms, corresponding to the interaction term *subway* × *frdc*. As observed, there is no significant difference in the innovation index between the treatment group and the control group before the impact, regardless of whether it involves the HSR-R&D center foreign firms or the subway-R&D center foreign firms combination. This lack of significant pre-impact differences indicates that there is no notable pre-trend before the shock, thereby satisfying the parallel trends assumption. For the HSR opening and the entry of R&D center foreign firms, a significant positive effect emerges in the fourth period after the shock, with a very significant and large positive effect appearing in the fifth period. Similarly, for the subway opening and the entry of R&D center foreign firms, a significant positive effect is observed in the third period after the shock, with a very significant and large positive effect also appearing in the fifth period. In summary, the DID model established for the HSR opening and the entry of R&D center foreign firms, as well as for the subway opening and the entry of R&D center foreign firms, satisfies the parallel trends assumption. There is no significant pre-trend before the shock, confirming the validity of the causal identification premise in the econometric model.

4.3. Placebo test

A potential explanation for the significant innovation complementarity effects observed between the opening of HSR and the entry of R&D center foreign firms, as well as between the opening of subway and the entry of R&D center foreign firms, could be that these effects are merely coincidental. For instance, cities where HSR or subway opened and R&D center foreign firms entered simultaneously might have experienced a coincidental increase in innovation levels during the study period due to some unobservable factors. In other words, the significant innovation complementarity effects between HSR and R&D center foreign firms, as well as between subway and R&D center foreign firms, may not be causal but rather the result of capturing some coincidental occurrence. To rule out this possibility, this study conducts a placebo test. Specifically, the two key treatment states (i.e., the simultaneous opening of HSR and entry of R&D center foreign firms, as well as the simultaneous opening of subway and entry of R&D center foreign firms) are randomly assigned at the city-industry level, and regressions are performed. This process is repeated 500 times, and the density distribution of the 500 regression coefficients is plotted. If the significant innovation complementarity effects between HSR and R&D center foreign firms, as well as between subway and R&D center foreign firms, found in this study are not driven by coincidence, then the vast majority of placebo regression coefficients should be smaller than the baseline regression coefficients.

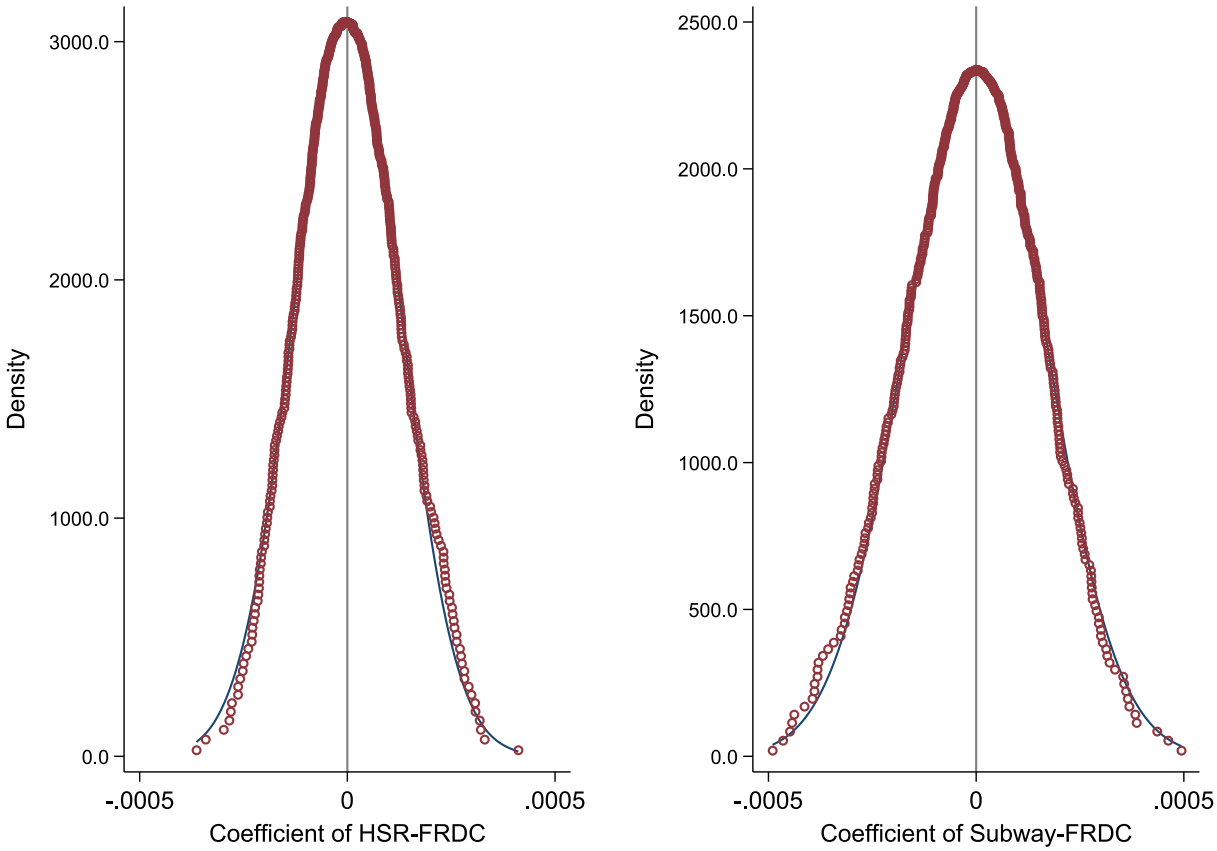

Fig. 3. Placebo test results.
Note: Fig. 3 illustrates the density distribution of the baseline regression coefficients from 500 random assignments of the simultaneous opening of HSR and entry of R&D center foreign firms, as well as the simultaneous opening of subway and entry of R&D center foreign firms.

Fig. 3 presents the density distributions of the placebo regression coefficients for the two key interaction terms: the left panel shows the results for HSR openings and R&D center foreign firms entry, while the right panel shows the results for subway openings and R&D center foreign firms entry. As seen, the regression coefficients from the 500 random assignments are approximately normally distributed around zero, regardless of whether it is for HSR or subway openings and R&D center foreign firms entry. Moreover, all placebo regression coefficients are very small, falling within the range of [−0.0005, 0.0005], and are smaller than the corresponding baseline regression coefficients (the core estimates in Column (4) of Table 2, which are 0.0477 and 0.0581, respectively). This suggests that the significant innovation complementarity effects found between HSR and R&D center foreign firms, as well as between subway and R&D center foreign firms, are indeed causal and not driven by coincidence.

4.4. Robustness checks

This study conducted robustness checks, including sample adjustments, changes in the study period, replacement of key variables, and consideration of stricter clustering levels and fixed effects. Appendix Table C2 reports the first part of the robustness test results. Firstly, column (1) of Appendix Table C2 presents the regression results after performing a 1 % winsorization at both ends of the dependent variable to exclude the influence of outliers. Secondly, the baseline regression restricts the study to municipalities directly under the central government and prefecture-level cities, excluding other prefecture-level administrative units (such as regions, autonomous prefectures, and leagues). Column (2) includes all prefecture-level administrative units. Thirdly, the baseline regression study period is from 2004 to 2020. To eliminate the impact of the 2008 global financial crisis, column (3) narrows the study period to 2008–2020. Fourthly, column (4) extends the baseline regression sample period to 2001–2020, showing that the results are robust. Fifthly, column (5) excludes the samples from the four municipalities directly under the central government and the five remote provinces (Inner Mongolia, Qinghai, Xinjiang, Tibet, and Hainan) to account for significant differences in HSR construction, subway construction, the attractiveness of R&D center foreign firms, and innovation levels between these regions. Sixthly, column (6) controls for more stringent fixed effects by replacing the gb4 industry-year fixed effects with province-gb4 industry-year fixed effects, thereby controlling for all time-varying characteristics at the province-gb4 industry level.

Appendix Table C3 presents the second part of the robustness checks. Firstly, one of the core interaction term main effects in the baseline regression is the binary variable indicating the presence or absence of R&D center foreign firms at the city-gb2 industry level, which examines the extensive marginal effect of R&D center foreign firms entry. Column (1) of Appendix Table C3 replaces this binary variable with the logarithm of the number of R&D center foreign firms at the city-gb2 industry level, exploring the intensive marginal effect of R&D center foreign firms entry. Secondly, the dependent variable in the baseline regression is the logarithm of the innovation index at the city-gb4 industry level; however, column (2) instead standardizes the original innovation index without logarithmic transformation. Thirdly, column (3) replaces the logarithmic transformation of the innovation index at the city-gb4 industry level with a hyperbolic sine transformation. It is evident that the coefficients of the core interaction terms are robust to different methods of processing the dependent variable. Fourthly, columns (4)–(6) consider more stringent clustering levels. Column (4) clusters at the more stringent city level, and both core interaction term coefficients remain significant at the 1 % level. Similarly, Column (5) clusters at the more stringent gb2 industry level, and both core interaction term coefficients remain significant at the 1 % level. Column (6) clusters at the more stringent provincial level, with both core interaction term coefficients still significant at the 1 % level. Therefore, the core effects in this study are robust to more stringent clustering levels.

In Section 3.3, this study finds significant differences between the treatment and control groups in terms of the innovation index and city market size. This section demonstrates that the results remain robust after controlling for these differences. The regressions based on Eq. (2) incorporate additional controls for the logarithm of the initial city-gb4 industry innovation index (*lninnovation*2003) and its interaction with year fixed effects, allowing the impact of initial city-industry innovation levels to vary across years. Furthermore, the logarithm of the city's initial GDP (*lngdp*2003) and total population (*lnpop*2003) and their interactions with gb4 industry-year fixed effects are included, permitting the influence of initial market size to differ across industries and years. The initial year corresponds to 2003, the year prior to the study period. Results in Appendix Table C4 show that the core estimated effects remain robust under these additional controls.

Existing research highlights potential biases in traditional two-way fixed effects (TWFE) estimators in staggered difference-in-differences models due to heterogeneous treatment effects (Callaway & Sant'Anna, 2021; De Chaisemartin & d'Haultfoeuille, 2020; Goodman-Bacon, 2021; Sun & Abraham, 2021). Firstly, this study applies the Goodman-Bacon decomposition method (Goodman-Bacon, 2021) to diagnose potential biases arising from heterogeneous treatment effects. Appendix Tables C4 and C5 present the decomposition results for the core explanatory variable in Eq. (1)—the binary variable representing R&D center foreign firms entry (*frdc*). Appendix Table C6 provides decomposition results for the interaction terms in Eq. (2): $hsr \times frdc$ (interaction of HSR and R&D center foreign firms entry) and $subway \times frdc$ (interaction of subway and R&D center foreign firms entry). The decomposition results confirm that biases due to heterogeneous treatment effects are minimal in both Eqs. (1) and (2). Secondly, this study also employs the heterogeneous treatment effect estimation method proposed by Callaway and Sant'Anna (2021). The corresponding results are shown in Appendix Table C7. The estimated coefficients for the heterogeneous treatment effects are significantly positive, further confirming the robustness of the findings.

5. Mechanism analysis

The theoretical analysis suggests that in the presence of R&D center foreign firms entry, HSR facilitates interregional exchanges, helping local firms access external markets and relax market access restrictions. Meanwhile, subway promotes intra-regional

exchanges, enabling local firms to approach R&D center foreign firms and ease technology acquisition restrictions. Thus, both types of transportation infrastructure amplify the technological spillover effects of R&D center foreign firms.

If the core mechanism proposed in this study holds, cities that simultaneously possess HSR, subway, and R&D center foreign firms should exhibit stronger innovation responses. These cities can simultaneously access external markets and R&D center foreign firms, increasing potential innovation benefits and reducing potential innovation costs. If this hypothesis is valid, the triple interaction term composed of HSR (*hsr*), subway (*subway*), and R&D center foreign firms entry (*frdc*) should have a positive coefficient. Table 2 reports the regression results, where the control variables and fixed effects are identical to those in Table 1, for comparability, with a focus on the triple interaction term ($hsr \times subway \times frdc$).

The regression results in Table 2 indicate that, the estimated coefficient of the triple interaction term is consistently and significantly positive. This finding suggests that HSR and subway jointly amplify the technological spillover effects of R&D center foreign firms. The results further support the study's core argument: cities with both HSR and subway experience simultaneous relaxation of market access restrictions and technology acquisition restrictions, thereby stimulating innovation activities more strongly. This leads to the observed compounding effect of HSR and subway in amplifying the technological spillover of R&D center foreign firms.

5.1. High speed rail, external market size and market access restrictions

This study finds that HSR and subway can jointly amplify the technological spillover effects of R&D center foreign firms. The coefficient of the triple interaction term $hsr \times subway \times frdc$—comprising the HSR binary variable, the subway binary variable, and the binary variable for R&D center foreign firms entry—is consistently and significantly positive. If the proposed mechanism of HSR is valid—namely, that HSR facilitates interregional exchanges, helping local firms access external markets and easing market access restrictions—then the following observation should hold: if the HSR binary variable (*hsr*) in the triple interaction term is replaced with an external market size indicator, the coefficient of the resulting triple interaction term should also remain significantly positive. This would further confirm that the amplification effect is driven by improved access to external markets facilitated by HSR. To construct the potential market size that a city faces, this paper follows the approach of Harris (1954):

$$size_{it} = (index_{it}/distance_{ii}) + \sum_{i \neq j} (index_{jt}/distance_{ij}) \tag{4}$$

$$size_{it} = \sum_{i \neq j} (index_{jt}/distance_{ij}) \tag{5}$$

In the eq. (4), the variable $size_{it}$ represents the national market size indicator that city i faces in year t, while $index_{it}$ represents the market size indicator for city i in year t. The term $distance_{ii}$ refers to the internal distance within city i, calculated as $distance_{ii} = (2/3)\sqrt{area_i/\pi}$, where $area_i$ is the area of city i. The variable $index_{jt}$ denotes the market size indicator of city j in year t, and $distance_{ij}$ is the bilateral distance between city i and city j. Therefore, $size_{it}$ essentially represents the weighted market size faced by city i from all other cities (including itself), with the weights determined by the distance between city i and other cities (where the internal distance applies for the city i itself). Thus, $size_{it}$ measures the national market size faced by city i. When calculating the national market size, the corresponding *index* for other cities is the same as the index used for city i itself.

Table 3
Mechanism analysis (I).

Dependent variable: ln*innovation*	(1)	(2)	(3)	(4)	(5)	(6)
Interaction variable △	National market (including the city itself)			National market (excluding the city itself)		
	GDP	wholesale retail	consumer retail	GDP	wholesale retail	consumer retail
△ × *subway*×*frdc*	0.210***	0.135***	0.199***	0.203***	0.136***	0.188***
	(0.0263)	(0.0163)	(0.0235)	(0.0234)	(0.0153)	(0.0211)
subway×*frdc*	−0.118***	−0.0951***	−0.122***	−0.0809***	−0.0692***	−0.0820***
	(0.0237)	(0.0198)	(0.0224)	(0.0181)	(0.0164)	(0.0176)
△ × *frdc*	0.0392***	0.0215***	0.0291***	0.0371***	0.0214***	0.0281***
	(0.00450)	(0.00246)	(0.00336)	(0.00443)	(0.00260)	(0.00346)
frdc	−0.00884	−0.00293	−0.00348	−0.00116	0.00395	0.00244
	(0.00759)	(0.00743)	(0.00735)	(0.00765)	(0.00755)	(0.00752)
ln*foreignnum*	4.17e-05	4.47e-05	6.40e-05	−1.43e-05	−9.05e-06	2.33e-05
	(0.000496)	(0.000495)	(0.000492)	(0.000500)	(0.000498)	(0.000495)
ln*rdcnum*	0.0130***	0.0130***	0.0127***	0.0130***	0.0129***	0.0128***
	(0.00192)	(0.00193)	(0.00192)	(0.00193)	(0.00194)	(0.00193)
City-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
City-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,965,808	3,965,808	3,965,808	3,965,808	3,965,808	3,965,808
R^2	0.780	0.780	0.781	0.780	0.780	0.780

Note: The values in parentheses represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively. The binary variables for subway opening *subway* and interaction variable △ and interaction term △ × *subway* are city-level time-varying variables, which are absorbed by the city-year fixed effects.

In this study, GDP is used as a proxy for market size, and a corresponding national market size measure is constructed. After taking the logarithm, this measure replaces the HSR binary variable (*hsr*) as the primary variable in the triple interaction term. The new triple interaction term is then incorporated into the regression model, with results presented in Column (1) of Table 3. The results show that, under GDP as a measure of market size, the coefficient of the new triple interaction term is significantly positive. This indicates that the national market size faced by the city, in combination with the subway, can jointly amplify the technological spillover effects of R&D center foreign firms. This finding aligns with the previous hypothesis that improved access to larger markets, facilitated by transportation infrastructure like subway, enhances the spillover effects.

Next, the study turns to two more specific market size proxy indicators: the total sales of goods in the wholesale and retail trade sector (hereafter referred to as "wholesale retail") and the total retail sales of consumer goods (hereafter referred to as "consumer retail"). The regression results for the corresponding national market size triple interaction terms are shown in columns (2) and (3) of Table 3. The qualitative conclusions remain unchanged. Columns (4) to (6) replicate the analyses of the previous columns, with the only difference being that the national market size now excludes the city's own size in its construction, meaning that eq. (4) now becomes eq. (5). The results in columns (4) to (6) are consistent with those in the previous columns.

5.2. Subway, internal space access and technology acquisition restrictions

Following a similar approach as outlined in Section 5.1, this study tests the hypothesis that subway enhance intraregional interaction, facilitating local firms' access to R&D center foreign firms and relaxing technological acquisition restrictions. If this proposed mechanism holds, replacing the subway binary variable (*subway*) in the triple interaction term with an internal spatial proximity measure should still yield a significantly positive coefficient.

Since it is not feasible to directly measure the spatial proximity between all local firms and R&D center foreign firms, this study uses the level of urban transportation development as a proxy for intracity spatial proximity. A more developed urban transportation system implies a smaller effective spatial distance between local firms and R&D center foreign firms. Specifically, the log of urban road area is used as a proxy for urban transportation development and replaces the subway binary variable (*subway*) in the triple interaction term. The regression results, presented in Column (1) of Table 4, show that the coefficient of the new triple interaction term is significantly positive. This finding suggests that urban transportation development, together with HSR, can jointly amplify the technological spillover effects of R&D center foreign firms. These results are consistent with the earlier hypothesis.

Subsequently, columns (2) and (3) use the number of taxis and buses as proxies for internal transportation development, respectively, taking the logarithm and constructing triple interaction terms. The coefficients of these triple interaction terms are also significantly positive, further confirming the previous conjecture. Columns (4) to (6) follow a similar approach as the previous columns but use administrative area or total population as the denominator to construct corresponding density or per capita indicators instead of total indicators (also taking logarithms): Column (4) is road area density, defined as the ratio of road area to administrative area; Column (5) is per capita taxis, defined as the ratio of the number of taxis to the total population; and Column (6) is per capita buses, defined as the ratio of the number of buses to the total population. The coefficients of the triple interaction terms are significantly positive when using road density, per capita taxis, and per capita buses as proxies for internal transportation level, indicating robustness.

Table 4
Mechanism analysis (II).

Dependent variable: ln*innovation*	(1)	(2)	(3)	(4)	(5)	(6)
Interaction variable △	Total			Density		
	urban road area	Taxi	Bus	urban road area	Taxi	Bus
△ × *hsr*×*frdc*	0.0735***	0.0466***	0.0649***	0.0471***	0.0415***	0.0662***
	(0.00902)	(0.00606)	(0.00763)	(0.00951)	(0.00662)	(0.00868)
hsr×*frdc*	−0.0502***	0.0105**	−0.0473***	−0.0241*	0.0440***	−8.89e-06
	(0.00875)	(0.00430)	(0.00940)	(0.0133)	(0.00413)	(0.00538)
△ × *frdc*	−0.00238	−0.00294	0.00259	−0.00692	−0.0311***	−0.0138**
	(0.00830)	(0.00858)	(0.00689)	(0.00818)	(0.00769)	(0.00588)
frdc	−0.00907	−0.0143***	−0.0158**	−0.00180	−0.00826	−0.00946
	(0.00709)	(0.00525)	(0.00641)	(0.0112)	(0.00624)	(0.00585)
ln*foreignnum*	−4.04e-05	−0.000122	8.46e-05	−6.18e-05	−0.000301	−0.000192
	(0.000506)	(0.000507)	(0.000494)	(0.000505)	(0.000525)	(0.000515)
ln*rdcnum*	0.0120***	0.0130***	0.0128***	0.0131***	0.0135***	0.0135***
	(0.00197)	(0.00199)	(0.00195)	(0.00197)	(0.00201)	(0.00198)
City-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
City-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-gb4 industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,890,292	3,907,344	3,908,156	3,887,044	3,907,344	3,908,156
R^2	0.782	0.775	0.778	0.780	0.775	0.774

Note: The values in parentheses represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively. The binary variables for HSR opening *hsr* and interaction variable △ and interaction term △ × *hsr* are city-level time-varying variables, which are absorbed by the city-year fixed effects.

It is important to note that the enhancement effect of transportation infrastructure to R&D center foreign firms might also stem from their joint promotion of local economic growth (Duranton & Turner, 2012). Additionally, transportation infrastructure might attract inventors, thereby generating innovation spillover effects (Agrawal & Goldfarb, 2008; Akcigit, Caicedo, Miguelez, Stantcheva, & Sterzi, 2018; Bell, Chetty, Jaravel, Petkova, & Van Reenen, 2019), which in turn enhance the technological spillover effects of R&D center foreign firms. Although this study cannot entirely rule out these possibilities, the core effects remain significantly positive even after controlling for the stock of R&D center firms and foreign firms at the city-gb2 industry level, as well as city-year fixed effects. Thus, the competitive explanations of economic growth and innovation spillover effects can be partially dismissed, further reinforcing the core mechanism proposed in this study: HSR loosens the market access restrictions for local firms by promoting inter-regional exchanges, while the subway loosens the technology acquisition restrictions for local firms provided by R&D center foreign firms through enhanced intra-city exchanges.

6. Further analysis

6.1. Regional market system and legal environment heterogeneity

This paper has demonstrated through both theoretical and empirical analyses that there is an enhancement effect of transportation infrastructure to R&D center foreign firms. Specifically, transportation infrastructure facilitates the technological spillovers from R&D center foreign firms. The core mechanism lies in the fact that R&D center foreign firms bring technological spillovers, with HSR loosening the market access restrictions for local firms and subway loosening the technology acquisition restrictions.

Indeed, merely loosening the market access restrictions and technology acquisition restrictions faced by entities may not be sufficient to significantly stimulate innovation. The reason is that innovation is an activity characterized by high uncertainty, significant investment, and substantial risk. Moreover, the outcomes of innovation are also vulnerable to unlawful infringement. Therefore, regional market institutions and the rule of law environment are fundamental considerations in the innovation decision-making process. As Acemoglu and Johnson (2005) pointed out, institutions, such as property rights protection and contract enforcement, play a crucial role in modern economic development. In regions with more robust market institutions and higher levels of the rule of law, entities can effectively mitigate various non-market costs during R&D and innovation activities and receive timely protection when facing unlawful infringement, thereby reducing the expected uncertainty of innovation activities. Following this analytical logic, it can be anticipated that in regions with well-developed market institutions and a sound legal environment, the enhancement effect of transportation infrastructure to R&D center foreign firms is stronger.

To test this hypothesis, this paper utilizes the provincial marketization index from Wang, Fan, and Hu (2019), with a particular focus on the sub-index of market intermediary organizations and legal institutional environment (hereafter referred to as intermediary law). The logarithm of the marketization index and the intermediary law sub-index is taken and then interacted with the core interaction terms to construct triple interaction terms. The regression results are shown in the first two columns of Table 5. It is evident that both the marketization index and the intermediary law sub-index yield positive and significant coefficients for the triple

Table 5
Regional market system and legal environment heterogeneity results.

Dependent variable: ln*innovation*	(1)	(2)	(3)	(4)
Interaction variable △	Marketization index	Intermediary law	IP model city	Number of lawyers per capita
△ × *hsr*×*frdc*	0.0789***	0.0487***	−0.00201	0.0703***
	(0.0176)	(0.00813)	(0.00456)	(0.0147)
△ × *subway*×*frdc*	0.465***	0.260***	0.163***	0.154***
	(0.0690)	(0.0350)	(0.0206)	(0.0277)
hsr×*frdc*	0.0153***	0.00652	0.0298***	0.00652
	(0.00478)	(0.00546)	(0.00464)	(0.00546)
subway×*frdc*	−0.0881***	−0.0510***	−0.0772***	−0.0510***
	(0.0233)	(0.0153)	(0.0173)	(0.0153)
△ × *frdc*	0.00972	−0.0201*	0.0125***	−0.0201*
	(0.0140)	(0.0114)	(0.00296)	(0.0114)
frdc	−0.0151*	−0.00805	−0.0125*	−0.00805
	(0.00780)	(0.00692)	(0.00684)	(0.00692)
ln*foreignnum*	6.10e-05	0.000290	7.39e-05	0.000290
	(0.000504)	(0.000449)	(0.000491)	(0.000449)
ln*rdcnum*	0.0136***	0.0114***	0.0127***	0.0114***
	(0.00197)	(0.00173)	(0.00191)	(0.00173)
City-gb4 industry fixed effects	Yes	Yes	Yes	Yes
City-year fixed effects	Yes	Yes	Yes	Yes
Year-gb4 industry fixed effects	Yes	Yes	Yes	Yes
Observations	3,966,620	3,961,748	3,968,244	3,673,488
R^2	0.776	0.778	0.779	0.782

Note: The values in parentheses represent city-gb2 industry-level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively. The binary variables for HSR opening *hsr*, subway opening *subway*, interaction variable △ and interaction term *hsr*×*subway*, △ × *hsr and* △ × *subway* are city (province)-level time-varying variables, which are absorbed by the city-year fixed effects.

interaction terms constructed with the binary variables for HSR opening and R&D center foreign firms entry, as well as with the binary variables for subway opening and R&D center foreign firms entry. This indicates that the higher the regional marketization level, the better the development of intermediary organizations, and the more robust the legal institutional environment, the stronger the enhancement effect of HSR (and subway) to R&D center foreign firms, and the greater the impact of HSR and subway on facilitating the technological spillovers from R&D center foreign firms.

Column (3) of Table 5 focuses on regional intellectual property (IP) protection. This study compiled a list of pilot cities designated as national IP model cities and constructed a binary variable to indicate whether a city is a designated IP model city. This binary variable was then interacted with the core interaction terms from the baseline regression. The regression results reveal a strong heterogeneous response of the enhancement effect of subway to R&D center foreign firms to the IP model city policy: compared to non-IP model cities, the enhancement effect of subway to R&D center foreign firms is significantly stronger in IP model cities. However, no significant heterogeneous response was found for the enhancement effect of HSR to R&D center foreign firms with respect to the IP model city policy. Although the coefficient of the triple interaction term is negative, its absolute effect is minimal. Column (4) uses the number of lawyers per capita at the province level as a proxy for the regional rule of law environment, and the logarithm of this variable was analyzed through a triple interaction term. The regression results indicate that both HSR and subway exhibit significant heterogeneous responses in their enhancement effects to R&D center foreign firms to the regional rule of law environment: the better the rule of law environment (as indicated by a higher number of lawyers per capita), the stronger the enhancement effect of HSR (and subway) to R&D center foreign firms. The analysis of the IP model city policy and the regional rule of law environment yields qualitatively similar results regarding heterogeneous effects, although the absolute magnitude and significance of the effects vary. This discrepancy may be because the IP model city policy and the number of lawyers per capita reflect different aspects of regional institutions, and further analysis will require subsequent research.

6.2. Regional spillover effect

The empirical study in this paper demonstrates a significant enhancement effect of transportation infrastructure to R&D center foreign firms at the city-gb2 industry level on innovation at the city-gb4 industry level. A natural question arises: Does the entry of R&D center foreign firms generate spillover effects, such as technology spillovers from neighboring regions with R&D center foreign firms? In the context of this study, the core issue is whether the complementary effect between R&D center foreign firms and transportation infrastructure at the city-gb4 industry level also applies to R&D center foreign firms in neighboring regions. To answer this question, the paper calculates the total number of R&D center foreign firms in all cities within a certain bilateral distance from the target city at the city-gb2 industry level. After taking the logarithm of this value, it replaces the baseline regression's core interaction term and the main R&D center foreign firm entry binary variable to investigate whether there is still a significant enhancement effect on the innovation level at the city-gb4 industry level of transportation infrastructure in the target city to R&D center foreign firms in neighboring city-gb2 industries. Importantly, the industry dimension is maintained, meaning that for a specific city-gb2 industry, only the total number of R&D center foreign firms in the same gb2 industry in surrounding cities is calculated. In this scenario, the paper estimates the enhancement effect of transportation infrastructure to R&D center foreign firms in all surrounding cities within a certain bilateral distance range on innovation at the city-gb4 industry level. The bilateral distances considered are (0,n], where $n = 100, 200, \ldots, 1000$, with the unit being kilometers (km). It should be noted that due to the potential endogeneity issues inherent in this regression

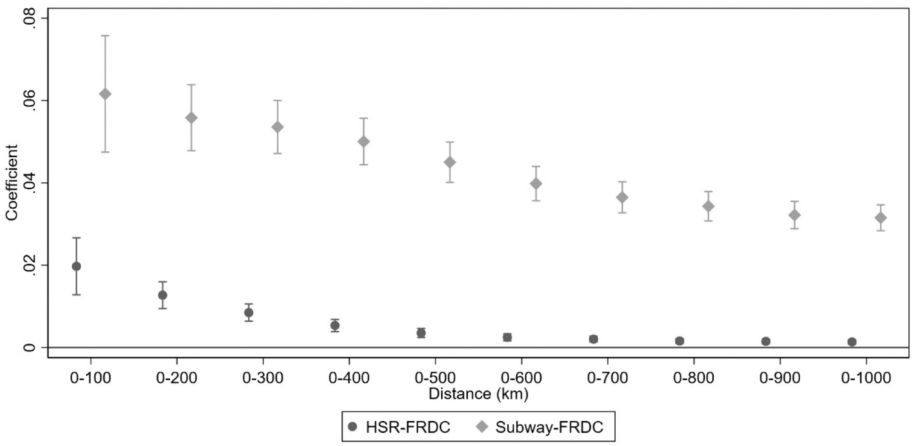

Fig. 4. Regional spillover effect results.
Note: Fig. 4 reports the estimated coefficients and corresponding 95 % confidence intervals for the interaction terms between the log of the number of R&D center foreign firms in all surrounding cities within a certain bilateral distance and the binary variables for high-speed rail (*hsr*) and subway (*subway*), while maintaining the industry dimension constant. The dots represent the coefficients for the interaction term involving *hsr*, and the diamonds represent those for *subway*. For each equal distance interval, the two key interaction terms are included in the same regression equation, consistent with the baseline regression specification.

design, the results can only be interpreted as correlations, serving as suggestive findings for future research.

Fig. 4 presents the regression coefficients of the core interaction terms and their 95 % confidence intervals. The dots show the regression results for the interaction terms corresponding to the HSR binary variable, while the rhombus show the results for the subway binary variable. The figure reveals that, for both HSR and subway, the enhancement effect of transportation infrastructure to neighboring cities' R&D center foreign firms gradually weakens as the bilateral distance increases. This finding is consistent with expectations, as knowledge exchange, technology diffusion, and personnel movement activities are significantly hindered by distance, aligning with the gravity model that predicts a decaying trend with increasing distance (Giroud et al., 2024).

6.3. Other findings

This section investigates whether the enhancement effect of transportation infrastructure to R&D center foreign firms contributes to the development of key industries in China. Appendix Table D1 focuses on two critical industries: high-tech industries and strategic emerging industries.

For high-tech industries, data were obtained from the National Bureau of Statistics' classifications of High-Tech Industries (Manufacturing) 2017 and High-Tech Industries (Service) 2018. These classifications provided a mapping of high-tech industries to the four-digit industry codes based on the 2017 version of the National Economy Industry Classification. Using this mapping, a binary variable was constructed at the gb4 industry level to indicate whether an industry is classified as high-tech. This binary variable was then interacted with the core interaction terms from the baseline regression to form a triple interaction term. The regression results are shown in column (1) of Appendix Table D1. The coefficient of the triple interaction term, which involves the high-tech industry binary variable, the HSR binary variable, and the R&D center foreign firm entry binary variable, is significantly positive. This suggests that the enhancement effect of HSR to R&D center foreign firms is stronger for high-tech industries compared to non-high-tech industries, meaning that HSR more effectively promotes technology spillovers from R&D center foreign firms in high-tech industries. However, the triple interaction term involving the subway binary variable is not significant, indicating that there is no significant industry heterogeneity in the enhancement effect of subway to R&D center foreign firms when comparing high-tech and non-high-tech industries.

For strategic emerging industries, data were obtained from the 2018 Strategic Emerging Industries Classification provided by the National Bureau of Statistics. This classification also provides a mapping of strategic emerging industries to the four-digit industry codes based on the 2017 version of the National Economy Industry Classification. Using this mapping, a binary variable was constructed at the gb4 industry level to indicate whether an industry is classified as a strategic emerging industry. This binary variable was then interacted with the core interaction terms from the baseline regression to form a triple interaction term. The regression results are shown in column (4) of Appendix Table D1. Unlike high-tech industries, the coefficients of the triple interaction terms for both HSR and subway are significantly positive. This indicates that the enhancement effect of transportation infrastructure to R&D center foreign firms is stronger for strategic emerging industries compared to non-strategic emerging industries. In other words, both HSR and subway more effectively promote technology spillovers from R&D center foreign firms in strategic emerging industries.

Therefore, this study provides evidence that the enhancement effect of transportation infrastructure to R&D center foreign firms contributes to the development of key industries in China, particularly by enhancing the innovation levels of high-tech industries and strategic emerging industries.

To date, all empirical analyses in this paper have confined the dependent variable to the innovation index at the city-gb4 industry level. Additionally, all analyses have identified the entry of R&D center foreign firms at the city-gb2 industry level. In this section, we further aggregate the dependent variable to higher levels to explore the aggregated effects of the enhancement effect of transportation infrastructure to R&D center foreign firms at different levels. This allows us to test the robustness of the core conclusions of this paper across different levels of the dependent variable. Simultaneously, this section identifies the entry of R&D center foreign firms at both the city-gb3 and city-gb4 industry levels, to verify the robustness of the core conclusions of this paper across different industry levels.

Appendix Table D2 reports the regression results when the dependent variable is aggregated from the city-gb4 industry level to the city-gb3 and city-gb2 industry levels, as well as when the entry of R&D center foreign firms is identified at the city-gb3 and city-gb4 industry levels. In summary, the core effect identified in this paper—namely, the enhancement effect of transportation infrastructure to R&D center foreign firms—is consistently positive and significant across various specifications, indicating robust results. The absolute effect varies depending on the level of industry aggregation used to identify the entry of R&D center foreign firms and the level of the dependent variable. This variation is attributed to the decay of technology spillover effects from R&D center foreign firms as industry distance increases (Arque-Castells & Spulber, 2022; Bloom, Schankerman, & Van Reenen, 2013; Myers & Lanahan, 2022), as well as the aggregated effects at different levels of analysis.

7. Conclusion

Attracting FDI is a crucial means for developing countries to absorb advanced international technological expertise. However, accurately identifying the technological spillover effects of FDI on host countries remains challenging, partly because FDI can also generate confounding effects, such as market competition. Moreover, even with significant FDI inflows, there is no guarantee of substantial technological spillover, as FDI requires sufficient external conditions to maximize its spillover potential. To address these issues, this paper utilizes the unique case of R&D center foreign firms and examines whether transportation infrastructure can facilitate technological spillovers from R&D center foreign firms. The theoretical analysis suggests that R&D center foreign firms bring technological spillovers, with HSR loosening the market access restrictions for local firms by promoting inter-regional interactions and

subway loosening the technology acquisition restrictions by enhancing intra-regional connectivity. Therefore, there is a significant enhancement effect of transportation infrastructure to R&D center foreign firms.

The primary findings and conclusions of this study are as follows: Firstly, the entry of R&D center foreign firms significantly enhances the innovation level within the same city-industry, and both HSR and subway can markedly amplify the technological spillover effects of R&D center foreign firms. Secondly, the core mechanisms lie in the following: HSR promotes interregional exchange, enabling local firms to access external markets and alleviating market access barriers; subway facilitate intraregional exchange, helping local firms gain proximity to R&D center foreign firms and reducing constraints on technology acquisition. Thirdly, the amplifying effect of transportation infrastructure on the technological spillover effects of R&D center foreign firms is stronger in regions with higher levels of marketization and better legal environments. Fourthly, R&D center foreign firms can generate cross-city technological spillover effects, and transportation infrastructure can enhance the spillover effects of such firms on surrounding city-industries. Fifthly, the amplifying effect of transportation infrastructure on the technological spillover effects of R&D center foreign firms is more pronounced in high-tech and strategic emerging industries.

This study offers two main insights for researchers and policymakers. First, traditional methods based on regional industry FDI scales, penetration rates, and FDI policy shocks struggle to distinguish between the technological spillover effects and market competition effects of FDI. Therefore, identifying unique FDI shocks that isolate market competition effects is necessary to obtain a pure estimate of technological spillover effects. The paper suggests that the approach based on the entry of R&D center foreign firms meets this criterion as a specific type of FDI entity. Second, society and government should recognize that FDI requires adequate external conditions to exert significant technological spillover effects, and favorable external conditions can fully harness FDI's potential for technological spillover in host countries. The study shows that both HSR and subway, despite their differing transport functions, can promote the technological spillover effects of R&D center foreign firms by leveraging their unique transportation advantages. Additionally, regional market institutions and legal environments can strengthen the enhancement effect of transportation infrastructure to R&D center foreign firms. Thus, both hard infrastructure and soft environmental factors are indispensable, and while advancing the construction of hard infrastructure such as transportation facilities, it is also crucial to foster soft environments like market institutions and legal frameworks.

Funding

This work was supported by the National Natural Science Foundation of China (72273068), and the National Natural Science Foundation of China (BEY010132).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chieco.2025.102367.

Data availability

Data will be made available on request.

References

Abebe, G., McMillan, M., & Serafinelli, M. (2022). Foreign direct investment and knowledge diffusion in poor locations. *Journal of Development Economics, 158*, Article 102926.

Abeberese, A. B., Barnwal, P., Chaurey, R., & Mukherjee, P. (2023). Democracy and firm productivity: Evidence from Indonesia. *Review of Economics and Statistics, 105*(3), 713–722.

Acemoglu, D., & Johnson, S. (2005). Unbundling institutions. *Journal of Political Economy, 113*(5), 949–995.

Acemoglu, D., & Linn, J. (2004). Market size in innovation: Theory and evidence from the pharmaceutical industry. *The Quarterly Journal of Economics, 119*(3), 1049–1090.

Agrawal, A., & Goldfarb, A. (2008). Restructuring research: Communication costs and the democratization of university innovation. *American Economic Review, 98*(4), 1578–1590.

Akcigit, U., Caicedo, S., Miguelez, E., Stantcheva, S., & Sterzi, V. (2018). Dancing with the stars: Innovation through interactions. In *NBER Working Paper, (No. 24466)*.

Alfaro-Urena, A., Manelici, I., & Vasquez, J. P. (2022). The effects of joining multinational supply chains: New evidence from firm-to-firm linkages. *The Quarterly Journal of Economics, 137*(3), 1495–1552.

Anderson, M. L. (2014). Subways, strikes, and slowdowns: The impacts of public transit on traffic congestion. *American Economic Review, 104*(9), 2763–2796.

Andersson, D., Berger, T., & Prawitz, E. (2023). Making a market: Infrastructure, integration, and the rise of innovation. *Review of Economics and Statistics, 105*(2), 258–274.

Arque-Castells, P., & Spulber, D. F. (2022). Measuring the private and social returns to R&D: Unintended spillovers versus technology markets. *Journal of Political Economy, 130*(7), 1860–1918.

Bell, A., Chetty, R., Jaravel, X., Petkova, N., & Van Reenen, J. (2019). Who becomes an inventor in America? The importance of exposure to innovation[J]. *The Quarterly Journal of Economics, 134*(2), 647–713.

Beraja, M., Kao, A., Yang, D. Y., & Yuchtman, N. (2023). AI-tocracy. *The Quarterly Journal of Economics, 138*(3), 1349–1402.

Beraja, M., Yang, D. Y., & Yuchtman, N. (2023). Data-intensive innovation and the state: Evidence from AI firms in China. *The Review of Economic Studies, 2023, 90*(4), 1701–1723.

Berger, T., & Prawitz, E. (2024). Collaboration and connectivity: Historical evidence from patent records. *Journal of Urban Economics, 139*, Article 103629.

Bloom, N., Schankerman, M., & Van Reenen, J. (2013). Identifying technology spillovers and product market rivalry. *Econometrica, 81*(4), 1347–1393.

Borusyak, K., & Hull, P. (2023). Nonrandom exposure to exogenous shocks. *Econometrica, 91*(6), 2155–2185.
Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics, 225*(2), 200–230.
Cantos, P., Gumbau-Albert, M., & Maudos, J. (2005). Transport infrastructures, spillover effects and regional growth: Evidence of the Spanish case. *Transport Reviews, 25*(1), 25–50.
Cao, J., Liu, X. C., Wang, Y., & Li, Q. Q. (2013). Accessibility impacts of China's high-speed rail network. *Journal of Transport Geography, 28*, 12–21.
Chen, Z., & Haynes, K. E. (2017). Impact of high-speed rail on regional economic disparity in China[J]. *Journal of Transport Geography, 65*, 80–91.
De Chaisemartin, C., & d'Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review, 110*(9), 2964–2996.
Donaldson, D. (2018). Railroads of the raj: Estimating the impact of transportation infrastructure. *American Economic Review, 108*(4–5), 899–934.
Donaldson, D., & Hornbeck, R. (2016). Railroads and American economic growth: A "market access" approach. *The Quarterly Journal of Economics, 131*(2), 799–858.
Duranton, G., & Turner, M. A. (2012). Urban growth and transportation. *Review of Economic Studies, 79*(4), 1407–1440.
Gebresilasse, M. (2023). Rural roads, agricultural extension, and productivity. *Journal of Development Economics, 162*, Article 103048.
Giroud, X., Lenzu, S., Maingi, Q., & Mueller, H. (2024). Propagation and amplification of local productivity spillovers. *Econometrica, 92*(5), 1589–1619.
Gong, R. K. (2023). The local technology spillovers of multinational firms. *Journal of International Economics, 144*, Article 103790.
Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics, 225*(2), 254–277.
Gu, Y., Jiang, C., Zhang, J., & Zou, B. (2021). Subways and road congestion[J]. *American Economic Journal: Applied Economics, 13*(2), 83–115.
Harris, C. D. (1954). The, market as a factor in the localization of industry in the United States. *Annals of the Association of American Geographers, 44*(4), 315–348.
Javorcik, B. S. (2004). Does foreign direct investment increase the productivity of domestic firms? In search of spillovers through backward linkages. *American Economic Review, 94*(3), 605–627.
Javorcik, B. S., & Spatareanu, M. (2008). To share or not to share: Does local participation matter for spillovers from foreign direct investment? *Journal of Development Economics, 85*(1–2), 194–217.
Jiao, J., Wang, J., Jin, F., & Dunford, M. (2014). Impacts on accessibility of China's present and future HSR network. *Journal of Transport Geography, 40*, 123–132.
Keller, W. (2004). International technology diffusion. *Journal of Economic Literature, 42*(3), 752–782.
Koh, Y., Li, J., & Xu, J. (2022). Subway, collaborative matching, and innovation. *Review of Economics and Statistics*, 1–45.
Kou, Z. L., & Liu, X. Y. (2017). *Report on innovation strength of cities and Industries in China*. Fudan Development Institute (in Chinese).
Liu, L., & Zhang, M. (2018). High-speed rail impacts on travel times, accessibility, and economic productivity: A benchmarking analysis in city-cluster regions of China. *Journal of Transport Geography, 73*, 25–40.
Liu, Q., Lu, R., Lu, Y., & Luong, T. A. (2021). Import competition and firm innovation: Evidence from China. *Journal of Development Economics, 151*, Article 102650.
Lu, Y., Tao, Z., & Zhu, L. (2017). Identifying FDI spillovers. *Journal of International Economics, 107*, 75–90.
Marinos, T., Belegri-Roboli, A., Michaelides, P. G., & Konstantakis, K.N. (2022). The spatial spillover effect of transport infrastructures in the Greek economy (2000−2013): A panel data analysis. *Research in Transportation Economics, 94*, Article 101179.
Myers, K. R., & Lanahan, L. (2022). Estimating spillovers from publicly funded R&D: Evidence from the US Department of Energy. *American Economic Review, 112*(7), 2393–2423.
Poole, J. P. (2013). Knowledge transfers from multinational to domestic firms: Evidence from worker mobility. *Review of Economics and Statistics, 95*(2), 393–406.
Setzler, B., & Tintelnot, F. (2021). The effects of foreign multinationals on workers and firms in the United States. *The Quarterly Journal of Economics, 136*(3), 1943–1991.
Severen, C. (2023). Commuting, labor, and housing market effects of mass transportation: Welfare and identification. *Review of Economics and Statistics, 105*(5), 1073–1091.
Shaw, S. L., Fang, Z., Lu, S., & Tao, R. (2014). Impacts of high speed rail on railroad network accessibility in China. *Journal of Transport Geography, 40*, 112–122.
Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics, 225*(2), 175–199.
Suri, T. (2011). Selection and comparative advantage in technology adoption. *Econometrica, 79*(1), 159–209.
Wang, Q., Liu, X., Zhang, F., & Hu, T. (2022). Subways and the diffusion of knowledge: Evidence from China. *China & World Economy, 30*(4), 60–99.
Wang, X. L., Fan, G., & Hu, L. P. (2019). *Marketization index of China's provinces*. China Social Science Academic Press (in Chinese).
Yang, J., Chen, S., Qin, P., Lu, F. W., & Liu, A. A. (2018). The effect of subway expansions on vehicle congestion: Evidence from Beijing. *Journal of Environmental Economics and Management, 88*, 114–133.
Yu, N., De Jong, M., Storm, S., & Mi, J. (2013). Spatial spillover effects of transport infrastructure: Evidence from Chinese regions. *Journal of Transport Geography, 28*, 56–66.